HARSHITH OBEROI

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SUMMARY

Data Scientist with 4+ years of experience across data analysis, machine learning, and applied analytics, supporting enterprise and financial services use cases. Proven in improving data reliability, reducing reporting turnaround, and delivering decision-ready analytics in regulated environments. Hands-on experience building applied ML and GenAI prototypes within analytics workflows. Strong in Python, SQL, feature engineering, and model evaluation, with practical exposure to cloud-based analytics on AWS. Experienced collaborating with cross-functional teams to translate ambiguous business problems into measurable analytics solutions.

SKILLS

Programming & Data Analysis: Python (Pandas, NumPy, SciPy), SQL (T-SQL, PL/SQL)
Machine Learning & Statistics: Regression, Classification, Ensemble Methods (Random Forest, XGBoost, LightGBM), Clustering (K-Means), Dimensionality Reduction (PCA), Feature Engineering, Model Evaluation (Precision, Recall, F1, ROC-AUC), A/B Testing
Time-Series & Forecasting: ARIMA, Exponential Smoothing, LSTM
Natural Language Processing: Text Preprocessing, Sentiment Analysis, Text Classification, Named Entity Recognition (NER), Embeddings, BERT (applied fine-tuning)
Generative AI: Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain (workflow orchestration), OpenAI APIs, Vector Embeddings (FAISS, managed services)
Data Engineering & Analytics Workflows: PySpark, Airflow (job orchestration), Databricks (analytics workflows), MLflow
Cloud & Platforms: AWS (S3, EC2, Glue, Lambda, SageMaker), Azure (ML Studio)
Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn
Databases: SQL Server, PostgreSQL, MySQL, MongoDB, Vector Databases

EXPERIENCE

Data Scientist – Analytics, State Street, USA — **Aug 2025 – Present**

- Built and maintained analytics-ready datasets supporting financial reporting, credit risk, and compliance teams used by 15+ stakeholders across risk and operations.
- Improved reporting turnaround from ~5 days to ~3 days by refactoring SQL logic, simplifying data dependencies, and reducing manual review steps.
- Collaborated with investment teams to build PySpark ETL pipelines on Databricks, integrating multi-source transaction data to support batch and near-real-time analytics.
- Executed automated data validation, reconciliation, and audit-trace checks to improve reporting reliability during regulated review cycles.
- Prototyped GenAI-assisted workflows using retrieval techniques (chunking, embeddings, FAISS) and LangChain to explore automation opportunities for document summarization and compliance review support.
- Partnered with analysts and risk stakeholders to standardize KPIs and reduce reliance on manual Excel-based workflows.
- Performed exploratory data analysis and feature engineering on credit and trading datasets to support risk modeling efforts.

Sr. Analyst, Capgemini, India (End client: Mercedes-Benz Research & Development) — **Mar 2021 – Jul 2023**

- Served as a senior analytics contributor supporting automotive R&D and manufacturing analytics initiatives.
- Managed telemetry and analytics pipelines processing 10M+ events per day, maintaining 99.7% system uptime in production environments.
- Reduced QA validation effort by ~60% by integrating ML-assisted test-case prioritization alongside rule-based checks in analytics workflows.
- Built time-series forecasting models to support EV battery health monitoring, warranty planning, and service decisioning.
- Developed Python-based automation tool to streamline engineering workflows, reducing manual process steps by over 75% and improving release efficiency.
- Designed analytics layers and dashboards consumed by 20+ global stakeholders for operational and product insights.
- Partnered with engineering, QA, and business teams to define KPIs, validate assumptions, and align analytics outputs with operational priorities.
- Built and maintained SQL- and Python-based analytics workflows, improving reporting consistency, data reliability, and turnaround time.

Data Analyst, Persistent Systems, India — **Jul 2019 – Feb 2021**

- Performed exploratory data analysis (EDA) and hypothesis testing on financial datasets using Python and R, identifying key trends that informed business strategies and improved forecasting accuracy by 15%.
- Generated SQL queries and Power BI dashboards to track KPIs, trends, and performance metrics.
- Developed basic regression models in Scikit-learn for risk assessment, contributing to early projects that reduced analytical errors by 20%.
- Optimized queries in PostgreSQL for reporting, reducing data retrieval time by 25% and supporting operational efficiency.
- Leveraged AWS services (S3, Glue, Lambda, EC2) to support data ingestion, transformation, and analytics workflows, enabling scalable reporting and analysis across large datasets.
- Supported ad-hoc analysis and investigative data requests with quick turnaround for stakeholders.

EDUCATION

Master's in Data Science, *Rutgers University, New Brunswick, New Jersey, USA* **Sep 2023 – May 2025**

Bachelor's in Information Technology, *Manipal University, Jaipur, Rajasthan, India* **Aug 2016 – Jul 2020**

CERTIFICATIONS

- **Oracle Certified Data Science Professional**
- **Microsoft Certified Azure Fundamentals**

PROJECTS

Lang2Query – Text-to-SQL System

- Built a retrieval-augmented Text-to-SQL system that translates natural-language questions into executable SQL over relational schemas for analytics use cases.
- Implemented schema chunking and embedding-based retrieval to improve query relevance and reduce invalid SQL generation.
- Evaluated generated queries against ground-truth outputs to analyze accuracy, error patterns, and failure modes in real-world analytical scenarios.

ADDITIONAL EXPERIENCE

Lecturer – Data Science, Rutgers University

- Taught *Introduction to Data Science* and *Databases for Data Science* courses.
- Led lectures, labs, and office hours for cohorts of ~50 students per class, covering Python, SQL, data modeling, and applied analytics concepts.